Unit 4 Comprehensive Review – Mixed

Directions: Show all work. Use exact values. Box or circle all answers.

1. Convert the explicit formula to a recursive formula: $a_n = 96(-\frac{1}{2})^{n-1}$. Include the first term.

2. A geometric sequence has $a_2 = 16$ and $a_5 = 2$. Find the common ratio and write an explicit formula for the sequence.

3. Solve $4 \cdot 3^x = 324$.

4. Find the common ratio and write the next three terms: $1024, 256, 64, 16, \ldots$

5. y-intercept = -5; common ratio = $\frac{1}{4}$. Write an explicit equation for the exponential function.

6. A quantity is 625 at the end of year 1 and increases by 20% each year. State the common ratio and find the value at the end of 4 years.

7. For $y = f(x) = (1500)(0.7)^x$, identify the y-intercept and common ratio. At what percentage is the function growing or decaying? Is it growth or decay?

8. Write an explicit formula for the geometric sequence $3, 12, 48, 192, ...$

9. Write a recursive formula to represent the sequence $4, 20, 100, 500, ...$

10. A game score has 64 when the situation begins and increases by 50% each period. Write an exponential equation to model the quantity at any given time. Find the value after 3 periods.

11. y-intercept = 9; common ratio = 2. Write an explicit equation for the exponential function.

12. A quantity has 3200 at the end of year 1 and decreases by 25% each year. Write a recursive formula.

13. Solve $2^{3x-2} = 32$.

14. Determine whether the sequence is geometric. If it is, find the common ratio: $2, -6, 18, -54, \ldots$

15. For $y = f(x) = (300)(1.12)^x$, identify the y-intercept and common ratio. At what percentage is the function growing or decaying? Is it growth or decay?

16. Convert the recursive formula to an explicit formula: $a_1 = 256,\ a_n = \frac{1}{4}a_{n-1}$.

17. A geometric sequence has $a_1 = -2$ and common ratio $r = -4$. Write the first 5 terms.

18. Solve $2^x = 128$.

19. Solve $5^x = \frac{1}{125}$.

20. The table shows the amount of medication at the end of each hour.

Hour	1	2	3	4
Milligrams	6400	1600	400	100

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

21. Write a recursive formula to represent the sequence $3, -6, 12, -24, \ldots$.

22. Point: $(2, 45)$; common ratio = 3. Write 2 different explicit equations that represent the same exponential function.

23. Write the first 6 terms of the sequence if $a_1 = 2,\ a_n = 4a_{n-1}$.

24. A machine value has 2500 when the situation begins and decreases by 20% each period. Write an exponential equation to model the quantity at any given time. Find the value after 3 periods.

25. A quantity is 1215 at the end of day 1 and loses one third of its value each day. State the common ratio and find the value at the end of 5 days.

26. Find the common ratio and write the next three terms: $2, 10, 50, 250, \ldots$

27. Write the first 5 terms of the sequence if $a_1 = 125,\ a_n = \frac{1}{5}a_{n-1}$.

28. Solve $4^x = \frac{1}{64}$.

29. Solve $3^{2x+1} = 243$.

30. A geometric sequence has $a_1 = 6$ and $r = 3$. Find a_5.

31. A geometric sequence has $a_3 = 28$ and $a_6 = 224$. Find the common ratio and write an explicit formula for the sequence.

32. Convert the explicit formula to a recursive formula: $a_n = 6(5)^{n-1}$. Include the first term.

33. Solve $5^{x+2} = 3125$.

34. A quantity is 4 at the end of round 1 and triples each round. find the value at the end of 6 rounds.

35. A geometric sequence has $a_1 = 1024$ and common ratio $r = \frac{1}{4}$. Write the first 5 terms.

36. A quantity has 7 at the end of round 1 and quadruples each round. Write a recursive formula.

37. Point: $(3, 8)$; common ratio = 2. Write 3 different explicit equations that represent the same exponential function.

38. Solve $5 \cdot 2^{x+1} = 320$.

39. The table shows the number of cells in a culture for hours 1 through 4.

Hour	1	2	3	4
Cells	9	27	81	243

Find the common ratio, write an explicit formula, and find the value at 6.

40. The table shows the mass of a sample for days 1 through 4.

Day	1	2	3	4
Mass	729	243	81	27

Find the common ratio, write an explicit formula, and find the value at 6.

41. A quantity is 2400 at the end of hour 1 and decreases by 50% each hour. State the common ratio and find the value at the end of 5 hours.

42. Solve $4^{5x-3} = 4^{x+9}$.

43. Write an explicit formula for the geometric sequence $5, -10, 20, -40, ...$

44. A geometric sequence has $a_1 = 160$ and $r = \frac{1}{2}$. Find a_6.

45. Solve $9^x = 729$.

46. A quantity has 40 at the end of hour 1 and increases by 50% each hour. Write a recursive formula.

47. A sequence has recursive formula $a_1 = 243,\ a_n = \frac{1}{3}a_{n-1}$. What is the common ratio? Write the first 5 terms.

48. Solve $3^{6x+1} = 3^{2x+13}$.

49. Write a recursive formula to represent the sequence $162, 54, 18, 6, ...$

50. Determine whether the sequence is geometric. If it is, find the common ratio: $5, 15, 45, 140, ...$

51. Point: $(2, 5)$; common ratio $= \frac{1}{2}$. Write 3 different explicit equations that represent the same exponential function.

52. Write the first 5 terms of the sequence if $a_1 = 6,\ a_n = -3a_{n-1}$.

53. Solve $2 \cdot 5^{2x} = 1250$.

54. Convert the recursive formula to an explicit formula: $a_1 = 7,\ a_n = 3a_{n-1}$.

55. The table shows the number of points earned at the end of each round.

Round	1	2	3	4
Points	8	24	72	216

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

56. Write an explicit formula for the geometric sequence $625, 125, 25, 5, \ldots$.